# Project Design Phase-II: Solution Requirements

| Field | Details |
|---|---|
| Date | 25 October 2026 |
| Team ID | N/A (Individual Project) |
| Project Name | Electric Vehicle Market Performance & Insights |
| Maximum Marks | 4 Marks |

## Functional Requirements

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task) | Priority |
|---|---|---|---|
| 1 | Data Access & Integration | Import and connect the global EV market dataset (CSV/Excel) containing sales volumes, pricing, and battery chemistries into Tableau. | High |
| 2 | Data Processing | Clean anomalies, handle null values, assign geographic roles to region fields, and correct data types for precise analysis. | High |
| 3 | Business Analysis | Use calculated fields to compute Executive KPIs: Total Sales, Average Price, Median Range, and OEM market share percentages. | High |
| 4 | Business Insights & Outputs | Build a 3x3 interactive dashboard and a 6-part Tableau Story mapping macro market trends to technical battery performance. | High |
| 5 | User Interaction & Analysis | Enable dashboard filters (Region, Battery Type, Segment) so stakeholders can drill down from global to OEM-specific data. | High |
| 6 | Data Update & Reporting | Allow the project to be saved locally (.twbx) and published online to Tableau Public for web-based stakeholder review. | Medium |

## Non-Functional Requirements

| NFR No. | Requirement | Description |
|---|---|---|
| NFR-1 | Usability | Clean 3x3 layout, professional fonts (e.g., Tableau Bold), and intuitive color coding for executive-level readability. |
| NFR-2 | Security | Published workbooks use open/public data, ensuring no sensitive PII or confidential corporate data is exposed on the web. |
| NFR-3 | Reliability | Calculations and aggregations consistently reflect accurate results based on the underlying dataset. |
| NFR-4 | Performance | The published dashboard and story points should render and filter within 3–5 seconds in a standard web browser. |
| NFR-5 | Availability | Hosted on Tableau Public, ensuring 24/7 web availability for users globally via a simple link. |
| NFR-6 | Scalability | Workbook architecture flexible enough to incorporate future datasets (e.g., next year's EV sales) with minimal redesign. |